

Answer Key — Variables — What You Change vs What You Observe

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1 **(b) changed (independent) variable** — the variable you set yourself is the changed/independent variable.
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- Q2 **(c) time to puff** — time to puff is the result you read off.
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- Q3 **(b) the thickness of each dough circle** — thickness must be kept the same so only temperature differs.
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- Q4 **(b) False** — changing two at once means you can't tell which caused the result.
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- Q5 **(a) Both A and R true, and R explains A** — keeping others constant is exactly why the result is fair.
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- Q6 **measured** — also called the dependent variable.
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- Q7 **qualitative** — a yes/no, word-based observation is qualitative.
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Short answer

- Q8 **Changed:** ramp length. **Measured:** distance the car rolls. **Kept same:** same car, same slope, same surface.
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- Q9 So that any change in the result can be linked to that one variable; otherwise we cannot tell which cause produced the effect.
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- Q10 **Qualitative:** colour of leaves (green/yellow). **Quantitative:** height in cm.
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Long / application

- Q11 **Changed:** type of flour. **Measured:** whether/how much it puffs (or time to puff). **Controlled:** dough thickness, dough size, oil temperature, way of dropping. Make two doughs differing only in flour, fry the same way, compare.
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- Q12 She changed two variables (temperature *and* stirring), so the faster dissolving could be due to stirring. Fix: stir both the same way, change only temperature.
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